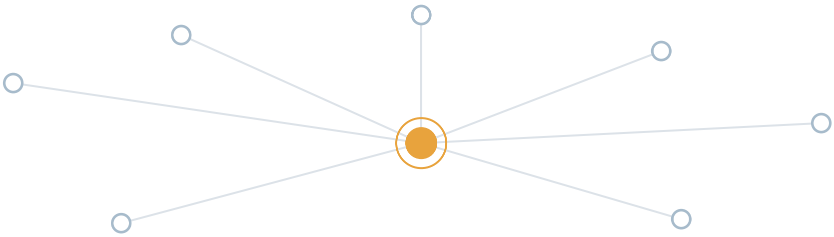


EXECUTIVE SUMMARY

AI Agents at the Border

The 50-Page Decision Brief

VALUE · AUTONOMY · SOVEREIGNTY · EXECUTION

THE OPERATING PRINCIPLE

Decisions may transfer within a signed mandate.
Accountability does not.

MAKSYM NECHEPURENKO

Dubai · 2026 · v1.6.0

A Decision Brief, Not a Substitute

This edition is the fast route through *AI Agents at the Border*. It is written for an executive team that needs a shared decision language, a bounded first portfolio, and a ninety-day operating plan before it needs the full evidential and implementation detail.

The brief preserves the book's central discipline: decisions may transfer within a signed mandate; accountability does not. It also preserves the evidence labels. A customs outcome, an adjacent-domain method, a consultancy-reported case, and a directional hypothesis are not interchangeable merely because they fit on the same slide.

Use the full book for legal applicability, procurement drafting, safety and security design, metrics, source attribution, and the complete Decision Toolkits. Use this brief to decide what the leadership team will do next.

Table of contents

How to Use This Brief

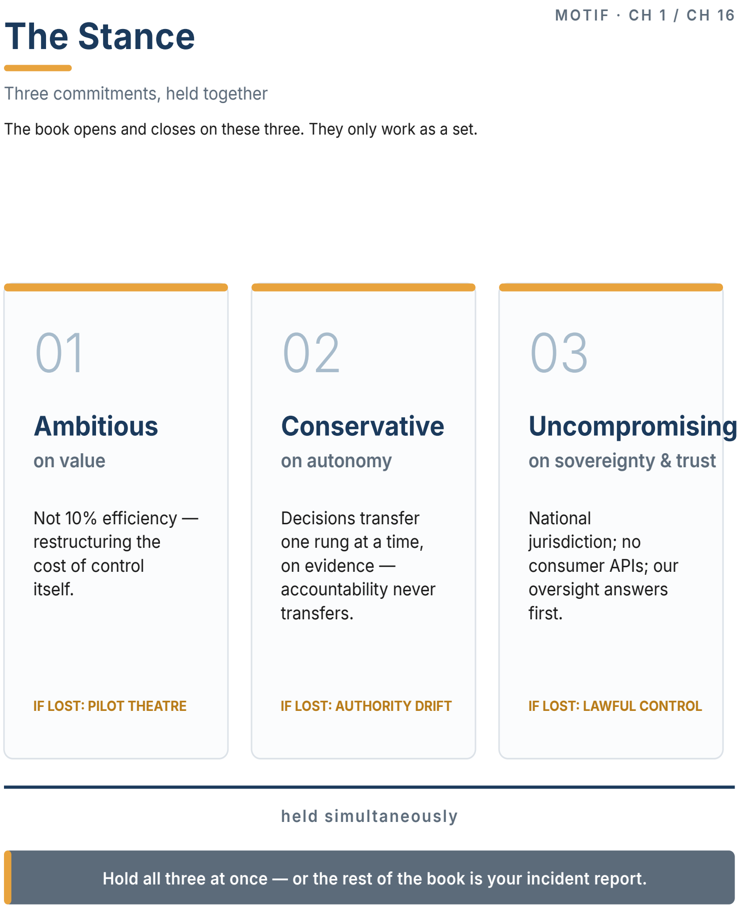
This brief is for a leadership team that has to turn *we should do something with AI* into a governed portfolio. It does not attempt to compress every clause, metric, case, and source from the full book. It preserves the decisions that determine whether the programme creates operating value or merely creates a new category of risk.

Read it once as a team. On the second pass, assign an owner to every decision in Chapter 8. Use the full book when the work reaches design, procurement, legal applicability, testing, or production approval.

The stance in one page

The book holds three commitments simultaneously.

Ambitious on value. The target is not a decorative chatbot or a ten percent improvement in a step that should not exist. It is a changed



The Stance

cost of control: more trade and more adaptive fraud handled without matching growth in officer effort, while facilitation becomes faster and more predictable.

Conservative on autonomy. A system earns authority one task and one rung at a time. A fluent model is not evidence that it may act. Promotion requires a local record of quality, exceptions, cost, security, human capacity, and reversibility.

Uncompromising on sovereignty and trust. Trade data, public authority, and the evidence needed to audit both remain under lawful administrative control. Consumer subscriptions, personal API keys, and unreviewed data paths are excluded territory. A commercial cloud region is not a sovereign deployment merely because it is nearby.

Four evidence labels

- **Verified:** a named administration or primary operating source identifies the technology and outcome.
- **Adjacent:** a neighboring domain supports a method or failure mode, not a customs forecast.
- **Practice-reported:** a public consultancy or delivery case; useful, attributed, and not independently verified.
- **Directional:** a hypothesis to test locally, not an observed result.

The label travels with the number. If it disappears when a claim enters a ministerial slide, the claim should disappear too.

> **LEGAL CUT-OFF · 10 JULY 2026.** This brief and the full book provide a dated design baseline, not legal advice or certification. Re-check every task with local counsel

before procurement, production approval, or an autonomy increase. The EU timetable stated here reflects the Digital Omnibus adopted on 29 June 2026; implementation material, national rules, and other jurisdictions continue to move.

INDEPENDENT WORK. The author writes in a personal capacity. This is not an Agility publication and does not represent or imply approval or endorsement by the author's employer, its clients, the WCO, or any cited administration. All cases and reported outcomes come from attributable public sources; none describes an Agility or client engagement.

The Eight Decisions

Start with the mandate, not the model

Customs leaders are not choosing whether AI exists. They are deciding where software may enter a public-authority workflow, what it may do there, which evidence earns more authority, and who remains answerable when it is wrong. Those are operating decisions. Model selection comes later.

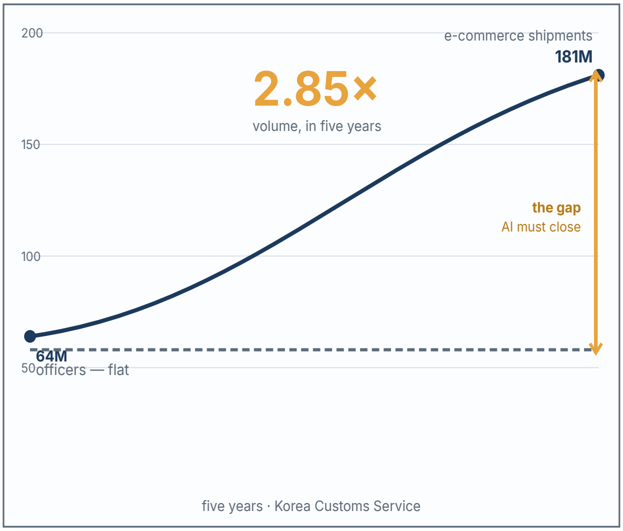
The arithmetic is already familiar: parcel and declaration volumes rise; fraud industrialises and adapts; experienced officers remain scarce; traders expect a digital service; government expects control and revenue without linear headcount. Classical rules and machine learning remain essential, but generative and agentic systems add a new option: software can read the unstructured record, assemble evidence, choose a next action, and carry work across systems. That last capability changes the governance question from *is the answer accurate?* to *what authority did the system exercise?*

CHAPTER 1

The Arithmetic

Why the mandate now needs more than headcount

Transaction volume grows far faster than any workforce plan. The widening gap is the case for AI.



Verified · administration-stated (KCS, WCO News 108).

Volume outgrows any workforce plan — the gap is the case for AI.

Figure 1.1 — The Arithmetic

The executive programme can be reduced to eight decisions.

1. **Name the operating constraint.** Choose a queue, service clock, control gap, or capacity ceiling. “Use AI” is not a problem statement.

2. **Choose the task and baseline.** Record current volume, time, quality, exceptions, cost, and responsible owner before changing the workflow.
3. **Classify autonomy and agency.** State who decides, which tools and data the software can reach, and which actions remain impossible.
4. **Set the sovereignty posture.** Decide lawful deployment, data paths, subprocessors, retention, and evidence control before the demo becomes architecture by accident.
5. **Build the value bridge.** Connect output to an operating action that changes capacity, facilitation, revenue protection, or control. Price the permanent human and assurance work.
6. **Choose what to own.** Buy commodity tools, partner for acceleration, and retain the institutional judgment, data meaning, evaluation, governance, and exit capability.
7. **Define the right to operate.** Pre-commit gates, a mandate, a safety case, a human route, security tests, demotion triggers, and fallback.
8. **Create a renewal rhythm.** Every task is periodically continued, changed, competed, demoted, or retired on evidence rather than momentum.

The first portfolio is deliberately uneven

An administration does not become “AI-ready” as a single event. It may be ready now for read-only document extraction, nearly ready for a grounded procedure assistant, and unready for a write-capable agent affecting a trader. That is normal. A useful portfolio contains three kinds of work:

- **value now:** bounded L1/L2 services using approved data and human confirmation;
- **foundations:** corpus stewardship, data contracts, identity, logging, evaluation, procurement, and supervisor capability;
- **options:** higher-autonomy tasks held in shadow mode until evidence and control capacity exist.

This avoids two symmetrical errors. The first is paralysis: treating every AI use as if it were an autonomous enforcement decision. The second is frontier theatre: funding an impressive agent before the administration can supply lawful data, absorb exceptions, observe trajectories, or stop the workflow safely.

The leadership test

Ask the proposed executive owner to complete one sentence:

> For [named task], the administration will allow [system] to operate at [autonomy rung] with [agency profile], because [local evidence] meets [pre-committed thresholds]; [named person] remains accountable, and the task will be demoted or stopped when [trigger] occurs.

If the sentence cannot be completed, the project is not ready for a model discussion. If it can, technology, legal, procurement, security, and operations finally have the same object to design.

Decision card

Before funding, record: named workflow · executive owner · current baseline · target service/control outcome · first authorised rung · agency profile · lawful deployment path · value-capture action · gate date · fallback · demotion trigger. One page is enough to expose whether the programme is buying an outcome or a story.

Agent, Autonomy, and Agency

The definition that belongs in the RFP

An AI agent is a software system that pursues an assigned goal by planning multi-step work and using tools—systems, databases, and documents—with a defined degree of autonomy and under defined human oversight.

Each phrase is a control. The goal bounds purpose. Planning creates the possibility of adaptive behaviour. Tools define reach. Autonomy allocates decision rights. Oversight retains human command. Remove any one and the buyer no longer knows what is being procured.

The bright-line test: can one decision-making component choose the next permitted action; can that choice change because of an intermediate result; and can it stop, re-plan, or escalate instead of following a fixed route? A pre-written chain remains a workflow or pipeline even when several steps use AI. Adaptive selection from bounded actions is agentic. This is the practical defence against agent washing.

Rules, machine learning, generative AI, and agents are not replacement generations. A robust customs workflow stacks them: rules enforce hard boundaries, ML supplies risk signals, GenAI reads and drafts, and an agent may orchestrate the allowed sequence. A vendor that proposes to replace hard constraints with a prompt has confused fluent language with control.

The Autonomy Ladder

L1 Assistant — Operator. The system drafts, retrieves, extracts, or summarises. A human checks every output before use. This is the default entry point for internal knowledge and document work.

L2 Copilot — Pilot. The system proposes an action inside the workflow; the officer confirms each consequential action. HS-code suggestion and document-completeness recommendations usually belong here.

L3 Supervised Agent — Supervisor. The system completes a bounded task across tools, while a human handles exceptions and retains the consequential decision. Case-file assembly is a strong candidate when the write path remains narrow and reversible.

L4 Autonomous Agent — Governor. The system decides and acts inside a machine-checkable mandate; humans audit outcomes and anomalies. It is defensible only for low-consequence, reversible,

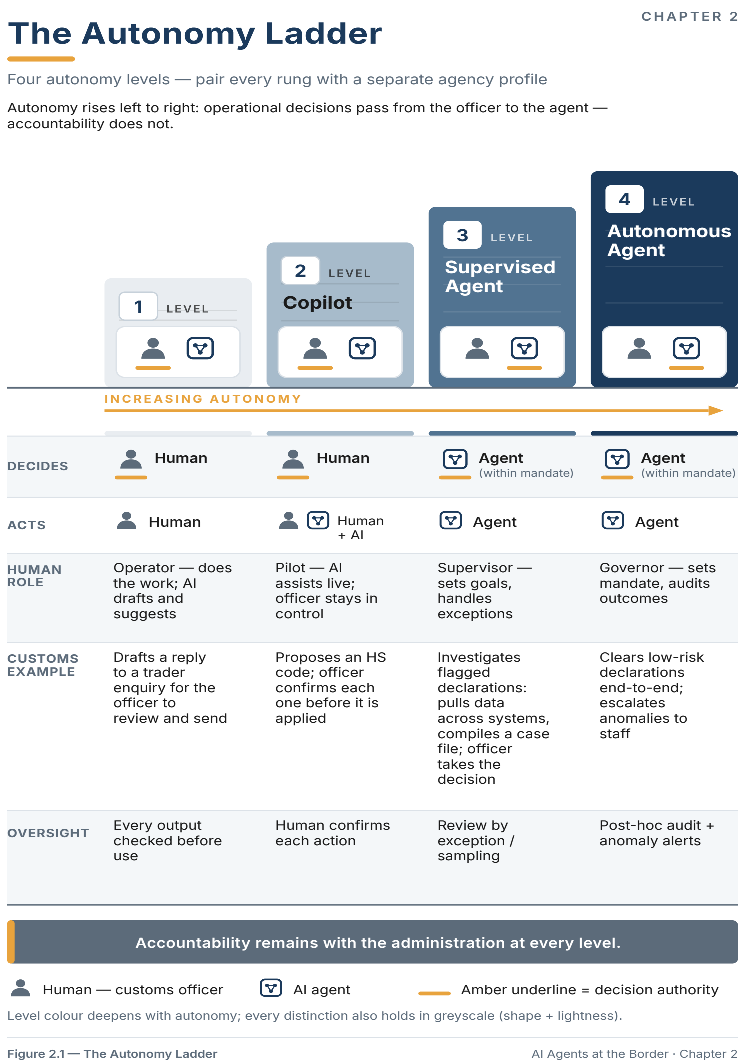


Figure 2.1 — The Autonomy Ladder

tightly bounded work with complete records and a tested fallback. Penalties and other materially adverse public-authority decisions do not earn this rung merely because a model performs well.

Promotion occurs one task and one rung at a time. A platform has no single autonomy level. On the same architecture, a search assistant may be L1, a classification suggestion L2, and a case-preparation loop L3.

Agency is the second dial

Autonomy asks who decides whether an action should occur. **Agency** asks what the software is technically able to touch: tools, data, permissions, write actions, communication channels, and transaction limits. The two must be recorded separately.

A read-only agent may adapt its research autonomously yet create little direct operational exposure. A copilot may require confirmation but still hold dangerous broad credentials. The control rule is **least agency**: grant only the specific capability needed for the task, use a distinct machine identity, deny everything else, and remove unused reach after observation.

The mandate card

Every deployed task needs a short, versioned mandate card:

- task trigger, start, end, and verifiable outcome;
- current autonomy rung and agency profile;
- permitted and prohibited tools, data, writes, and external messages;
- evidence shown to the officer and trajectory retained for audit;

- named accountable owner and human review route;
- quality, cost, safety, security, and queue thresholds;
- material-change and re-authorisation rules;
- demotion, suspension, fallback, and expiry.

No completed card means no authority to operate. Updating a prompt is an ordinary change only when it does not alter behaviour, evidence, data, tools, or authority. Otherwise it re-enters the gate.

Six questions for any "agent"

Task · rung and why not lower · agency and why each permission is needed · mandate and signer · reconstructable evidence · oversight and demotion. An answer that exists only in a vendor deck is not an operating control.

Value, Portfolio, and Readiness

The ladder of institutional readiness

The fastest credible route to agents begins below agents.

Document intelligence makes invoices, certificates, declarations, rulings, and correspondence machine-readable. **Grounded assistants** use approved corpora and return citations, corrections, and explicit uncertainty. **AI agents** then use those governed tools and evidence to close bounded loops. Skipping the lower rungs does not create speed; it pushes document, data, and verification defects into a system now able to act on them.

Start with a portfolio, not a single flagship. Score tasks on value, frequency, data readiness, reversibility, legal consequence, integration, human exception capacity, and security exposure. Set an autonomy ceiling before a vendor proposes one. The first agent should watch a defined queue, gather authorised evidence, prepare a cited recommendation, request a human decision, and write back

Figure 3.1 — The Maturity Ladder

only the approved result. Its most valuable early output is often the operational record: missing data, recurring exceptions, override reasons, unit cost, and useful tool permissions.

Four value channels

Efficiency: capacity absorbed, cycle time, handoffs removed, and evidence-to-decision latency. Korea Customs Service reports high-risk cargo analysis moving from roughly one hour to under one minute and estimates about USD 92 million annual savings while e-commerce volume expanded. This is administration-stated, verified customs evidence—and it reflects a decade of capability, not a ninety-day pilot.

Control: better targeting at a fixed inspection or analyst budget. Brazil's SISAM reported inspections roughly twenty times more effective than random selection. China reports more than 20,000 AI image-analysis detections since 2022 and recognition accuracy above 95 percent. Carry the named technology, period, and source with each number.

Revenue protection: changed assessments or prevented leakage. Public AI-attributable customs numbers remain scarce. Tax and VAT examples can show how to measure detection value, but they are adjacent evidence, not a customs forecast.

Facilitation: predictable clearance, fewer document cycles, quicker answers, and lower compliance effort. Do not attribute single-window or process-reengineering gains to AI. Name the precise step AI changes and measure it locally.

> ***ADJACENT EVIDENCE — HYPOTHESIS, NOT FORECAST.*** *Borrow a method or failure mode from tax, financial crime, logistics, or enterprise operations. Never borrow its benefit rate. Replace the percentage with a local baseline and representative pilot.*

The value bridge

The system output is not the benefit. A summary creates value only when it shortens a named officer step. A risk score creates value only when the inspection plan changes and the review burden remains manageable. An agent creates value only when completed work changes capacity, service, revenue, or control.

Build the economic record in this order:

1. baseline volume, time, quality, exceptions, and cost;
2. eligible share and realistic adoption ramp;
3. measured time or quality delta by evidence grade;
4. conversion action—what management actually changes;
5. permanent model, cloud, integration, data, security, oversight, and change costs;
6. downside provision and fallback;
7. sensitivity across volume, unit price, adoption, and exception rate.

Avoided headcount growth is often more credible than layoffs. Faster work has no cash value if staffing, service levels, or risk coverage never change. Name the benefit owner and the decision that captures it.

Readiness routes; it does not rank

Assess readiness per task and rung across leadership, legal/sovereignty, data, technology/operations, people, and governance. Demand artifacts, not confidence scores: named owner, approved corpus, data-flow map, deployment decision, identities, logs, evaluation set, supervisor, mandate, and gate.

An internal read-only search service can proceed at Rung 1/L1 with an approved corpus, citations, controlled retention, and human use even while the same administration remains unready for a trader-facing write agent. The weakest binding dimension sets the ceiling for that task; it does not freeze the entire portfolio.

The sovereignty decision

Choose on-premises, legally recognised government/sovereign cloud, or a hybrid by data class and workload. A nearby commercial region is not accreditation. Counsel and the competent security authority verify the provider, service, subprocessors, data path, terms, and purpose.

Inventory any existing consumer tools and personal API keys. Stop new sensitive uploads, provide a sanctioned synthetic/public-data sandbox, and migrate, contain, or retire each workflow against a date. A prohibition without a transition route is an invitation to shadow IT.

Buying Without Surrendering Judgment

Build, buy, partner

The useful boundary is not source code versus subscription. It is institutional judgment versus commodity capability.

Buy OCR, translation, infrastructure, identity, monitoring, generic workflow, and other mature layers when the market can supply them under lawful terms. **Partner** where the problem is strategic but engineering and model capability move too quickly to carry alone. **Build and retain** the administration's problem definition, data meaning, risk logic, evaluation, governance, operating ownership, and ability to exit.

Korea demonstrates what long-term internal capability can compound. It does not offer a template that can be bought quickly: its conditions include a decade of investment, national operational

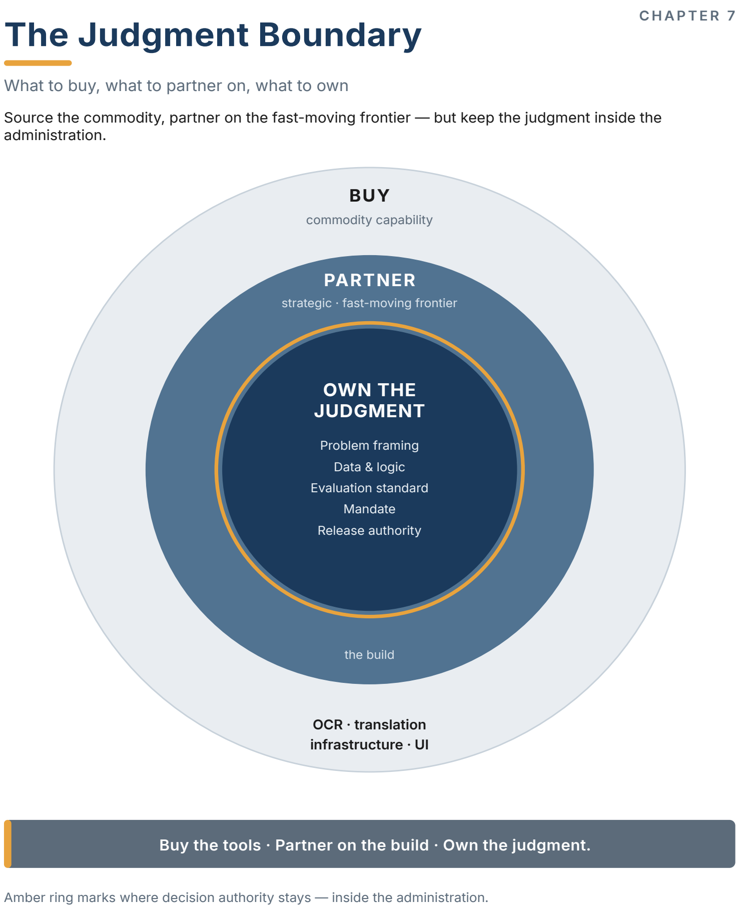


Figure 4.1 — The Judgment Boundary

data, a stable technical cadre, partnerships, and administrative continuity. The transferable principle is not “build everything.” It is: buy the tools, partner on the build, own the judgment.

The RFP that cannot buy the wrong thing

Specify the decision and operating boundary, not a fashionable solution. For every task require the bidder to answer:

- outcome, baseline, users, and exceptions;
- autonomy rung and why not lower;
- agency profile—tools, data, identities, permissions, writes;
- draft mandate, exclusions, human route, and signer;
- lawful deployment, region, subprocessors, retention, and training use;
- evaluation protocol, cost record, trajectory export, incident path, demotion, fallback, and exit.

The buyer owns the evaluation. A bidder may help design it but does not choose all test cases, define success after seeing the result, or keep the only copy of the evidence.

Use twenty cases correctly

Give the vendor twenty representative, unseen cases from the real workflow and observe the entire path: ingestion, evidence, intermediate choices, latency, retries, failure, human effort, and unit cost. Include ordinary messiness—languages, duplicates, missing pages, poor scans—not only adversarial curiosities.

This is qualitative pre-qualification. Twenty cases can reveal integration theatre, hidden manual work, missing citations, and a brittle demo. They cannot estimate a production error rate or establish safety. Formal acceptance uses a stratified, repeated, buyer-controlled evaluation, sealed holdout, hostile cases, baseline comparison, and pre-committed thresholds.

Contract the operating system

The critical clauses cover more than model performance:

Rights and evidence. The administration retains lawful use of its source and transformed data, annotations, feedback, evaluation sets, outputs, performance records, configurations, mappings, and documentation needed to audit and migrate. Vendor background IP can remain vendor IP without trapping the public record inside it.

Change. Model, prompt/policy, corpus, tool, identity, connector, region, and critical dependency changes are classified, tested, recorded, and re-authorised when material. "Continuous improvement" is not permission for silent authority expansion.

Operations. Service levels cover evidence freshness, exception queues, trajectory availability, containment, fallback, and change—not only platform uptime.

Exit. Require readable exports, documented interfaces and mappings, transition assistance, termination capacity, credential revocation, and a rehearsed fallback. Ask annually: if the contract ended in twelve months, what exactly could the administration take and operate?

Commercial transparency. Price inference, storage, integration, managed operations, human review, data work, security testing, change, and exit by task. A cheap model with an expensive exception queue is not a cheap service.

The evidence room

From month one, keep an administration-controlled record of mandate cards, architecture/data flows, component inventory, test sets/results, releases, trajectories, security findings, incidents, training, interfaces, runbooks, and contract evidence. Export it before final acceptance. This converts lock-in from a general fear into testable missing artifacts.

LOCAL-LAW ADAPTATION REQUIRED. *The full book's RFP skeleton is a control template, not procurement or contract advice. Local counsel adapts IP, audit, liability, records, security, and remedy clauses. When direct ownership or source-code access is unavailable or disproportionate, use a lawful functional substitute—tested escrow, readable exports, interfaces, independent assurance, transition assistance, and fallback.*

Govern the Task, Not the Label

A three-layer legal view

Every deployment sits inside three layers. **Binding law** covers data protection, secrecy, records, cybersecurity, administrative procedure, procurement, employment, and increasingly AI-specific duties. **Government policy and assurance** turns principles into hosting approvals, impact assessments, model registers, testing, and purchasing conditions. **Soft law and standards**—WCO, OECD, NIST, ISO/IEC and sector guidance—often becomes operationally hard when it enters a tender, audit, insurer, or ministerial approval.

The EU AI Act remains the most influential reference regime. Border management, migration/asylum, biometrics, and some law-enforcement uses can fall into Annex III high-risk categories. Following the Digital Omnibus adopted on 29 June 2026, the customs-relevant standalone high-risk date is 2 December 2027; embedded-product high-risk rules apply from 2 August 2028. Article 50 transparency solutions have a 2 December 2026 timetable. Application still depends on the exact task, deploying entity, people, purpose, and jurisdiction.

Other regimes do not collapse into an EU checklist. Data localisation, automated-decision rights, high-impact definitions, public-law duties, government cloud rules, and regulator guidance differ across the GCC, MENA, Asia, and trading partners. Build a strong common control base, then perform a named-task legal delta review.

> ***PORTABLE BASELINE, NOT COMPLIANCE GUARANTEE.*** *Human oversight, explainability, documentation, auditability, residency discipline, and standards alignment reduce redesign. They do not certify a system or make "meaningful human involvement," "high impact," or automated decision-making equivalent across jurisdictions. Local counsel confirms applicability and evidence.*

From requirement to control

Do not ask legal to approve "the AI programme." For each task, record: source and applicability · concrete obligation · operating control · evidence that the control works · owner · re-test trigger. A human- oversight duty becomes an interface, named role, escalation, override log, training, and capacity. A transfer restriction becomes an approved region, subprocessor list, encryption boundary, contract right, and test.

Maintain a quarterly regulatory watch list with official source, status, affected mandate cards, owner, next review, probability over 12–24 months, no-regret preparation, and irreversible commitment to avoid. A proposal is not enacted law; a watch list must not turn it into one.

Autonomy × agency

Oversight follows both the decision rung and technical reach. High agency with broad write tools may require stronger identity, approval, and monitoring even at L2. High autonomy with narrow, reversible, read-only reach may be manageable under sampling and anomaly review. Route by regulatory impact, affected-party proximity, reversibility, data sensitivity, agency, and human exception capacity.

Three human patterns remain useful:

- **human in the loop:** confirms each consequential action;
- **human on the loop:** supervises bounded execution and can intervene;
- **human in command:** sets mandates, resources, audit, promotion, demotion, and retirement.

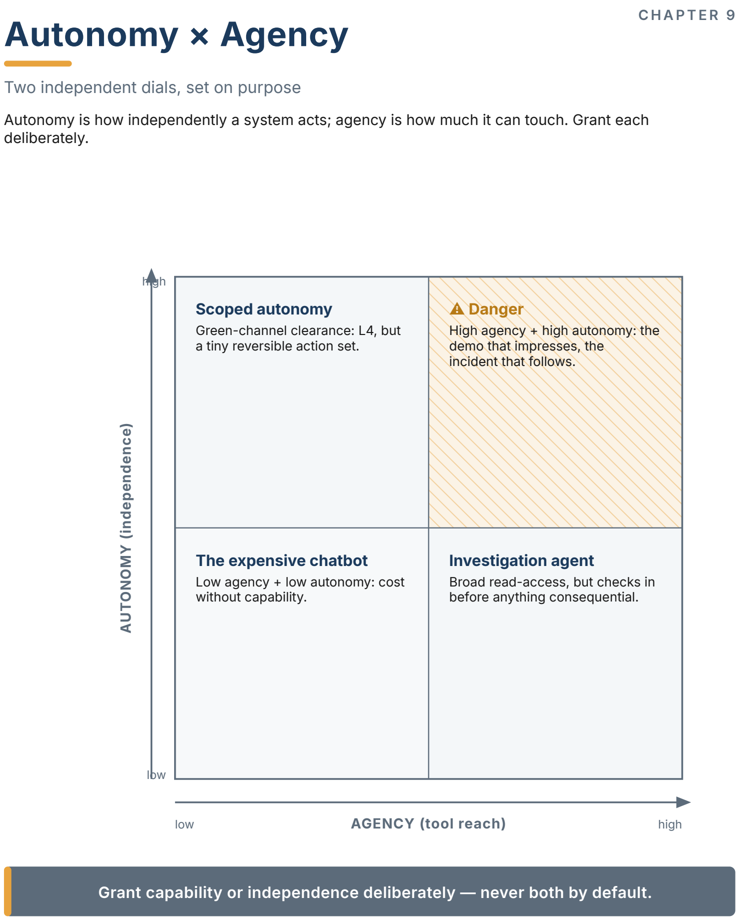


Figure 5.1 — Autonomy × Agency

“Human oversight” is not satisfied by a person theoretically able to watch an impossible queue. Measure queue size, exception age, comprehension, override quality, fatigue, and authority to stop.

The internal safety case

A safety case is the concise argument that a named task may operate at a specified rung with stated residual risk. It links the mandate, data flow, evaluation by relevant strata, hostile-input and tool-misuse tests, trajectory samples, human calibration, incident/fallback exercise, owner, and dated sign-off. Review it before pilot, autonomy expansion, material change, renewal, and after an incident.

In this book, *safety case* is an internal governance artifact. It is not a regulatory filing, conformity assessment, certification, or claim of equivalence with aviation, medical-device, nuclear, or other formal safety regimes. Use the locally required name while preserving the evidence logic.

Rights, fairness, and contestability

Where a workflow influences a trader, traveller, or employee, define what is communicated, what evidence can be seen, how a meaningful human review is requested, how the original output is isolated, who can correct the record, and how recurrent errors feed change control. Explanation is not a paragraph generated by the same model. It is the usable connection between source evidence, policy, outcome, and review.

Fairness testing follows the actual service: language, document quality, commodity, route, trader segment where lawful, exception type, and operational burden. Monitor both model error and who absorbs the delay.

Agentic security

Treat trader-supplied documents, email, web content, and external agent messages as hostile input. Separate the **content channel** from the **control channel**: untrusted text may inform evidence, but it cannot become instruction or expand permissions.

Require distinct agent identities, least agency, policy-enforced tool allowlists, argument validation, transaction limits, secrets outside prompts, immutable trajectories, anomaly detection, signed updates, component provenance, freeze/preserve/fallback runbooks, and repeated hostile-document tests. “Read only” is a claim until underlying scopes prove it.

Data as operating evidence

Agents need machine-consumable documents, governed knowledge, tool-ready interfaces, and feedback that records corrections rather than hiding them. Give each corpus and feed an owner, purpose, inclusion/exclusion rule, provenance, validity, freshness service level, quality thresholds, and failure response. A retrieval system with no source custodian is a demo with an expiry date.

Pilot, Production, and People

Pilot the operating system

A useful pilot tests more than whether a model can produce an answer. It tests whether the administration can supply lawful evidence, operate the human gate, observe behaviour, absorb exceptions, control cost, respond to failure, and decide against its own enthusiasm.

Use four stages.

Design and offline evaluation. Map the current workflow and baseline. Define the task, mandate, autonomy ceiling, agency profile, affected people, failure modes, test strata, thresholds, and fallback. Evaluate on buyer-controlled data with repeated trials and a failure taxonomy.

Shadow mode. Run against real or authorised representative traffic without affecting outcomes. Compare with the current path. Mea-

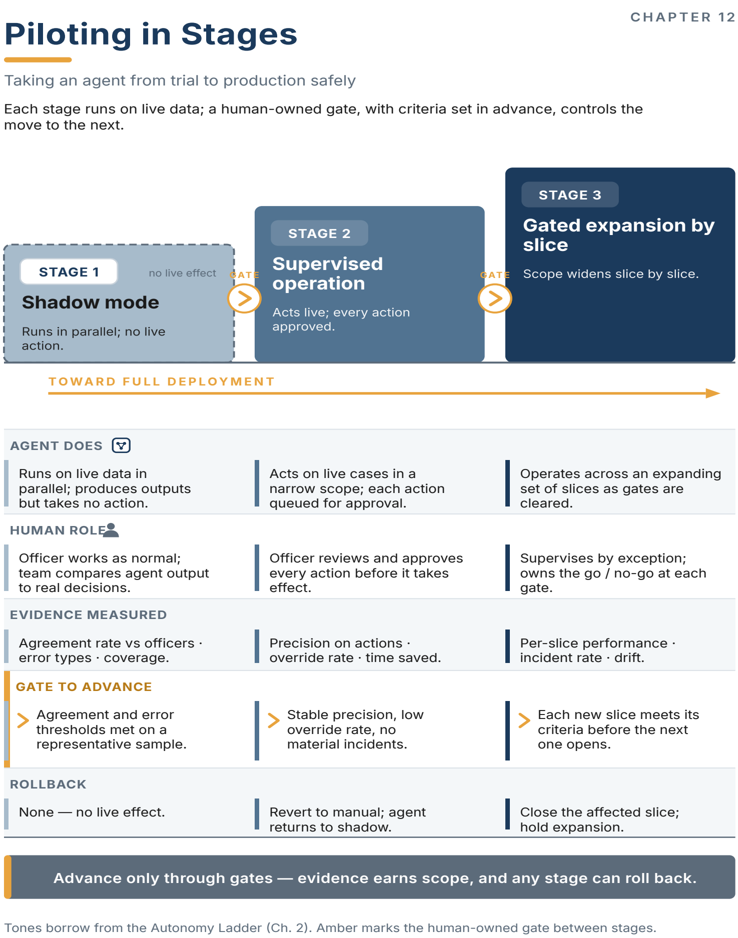


Figure 6.1 — Piloting in Stages

sure consistency, evidence quality, abstention, overrides, latency, unit cost, security, and the exception queue.

Controlled live pilot. Allow a narrow user group, limited volume, and reversible actions. Staff the human route as production work.

Exercise freeze, evidence preservation, containment, communication, and fallback.

Limited production. Approve only the tested mandate. Canary material changes, retain a holdout, sample trajectories, and publish a gate memo: promote, hold/tune, demote, or stop.

Reliability compounds across steps

Single-shot accuracy disguises workflow risk. If five dependent steps each succeed 95 percent of the time, the probability of an error-free chain is about 77 percent before correlations and adversarial conditions. Measure complete-case success, state transitions, retries, loops, tool errors, incorrect writes, and escalation—not only the final text.

Repeat identical and equivalent cases. Generative systems are stochastic; a result that passes once may fail under another run, model release, prompt, corpus, or tool state. Pre-commit the number of trials and acceptance rule.

The production control room

Join six views around the task: volume/service · quality/overrides · tools and trajectories · security/incidents · unit economics · mandate and owner. The meeting ends in a decision, not a dashboard: continue, tune, hold, demote, expand, compete, or retire.

Govern four change classes at minimum—model, prompt/policy, corpus, and tool list—and extend the register to identities, connectors, region, workflow, and critical dependencies. Every material change carries owner, test, canary/shadow path, communication, evidence-room update, rollback, and re-authorisation decision.

The second use case is the platform test. It should reuse identity, logging, evaluation, security, evidence, deployment, procurement, and control-room components. If almost nothing transfers, the first project built an implementation, not an institutional capability.

The workforce does not disappear

The officer role shifts with the ladder: Operator, Pilot, Supervisor, Governor. Higher autonomy usually reduces routine touches but increases exception complexity, calibration, sampling, incident work, and the need for people able to challenge both system and policy.

Use a hub-and-spoke model: a central AI/data team owns shared engineering, evaluation, security patterns, and standards; officer-analysts in operational units own domain meaning, exceptions, feedback, adoption, and service outcomes. Pair them. A central team without operations builds generic tools; isolated units duplicate platforms and controls.

Train by role. Users learn evidence limits, uncertainty, secure handling, and correction. Supervisors learn queue management, sampling, trajectory review, demotion, and incident escalation. Own-

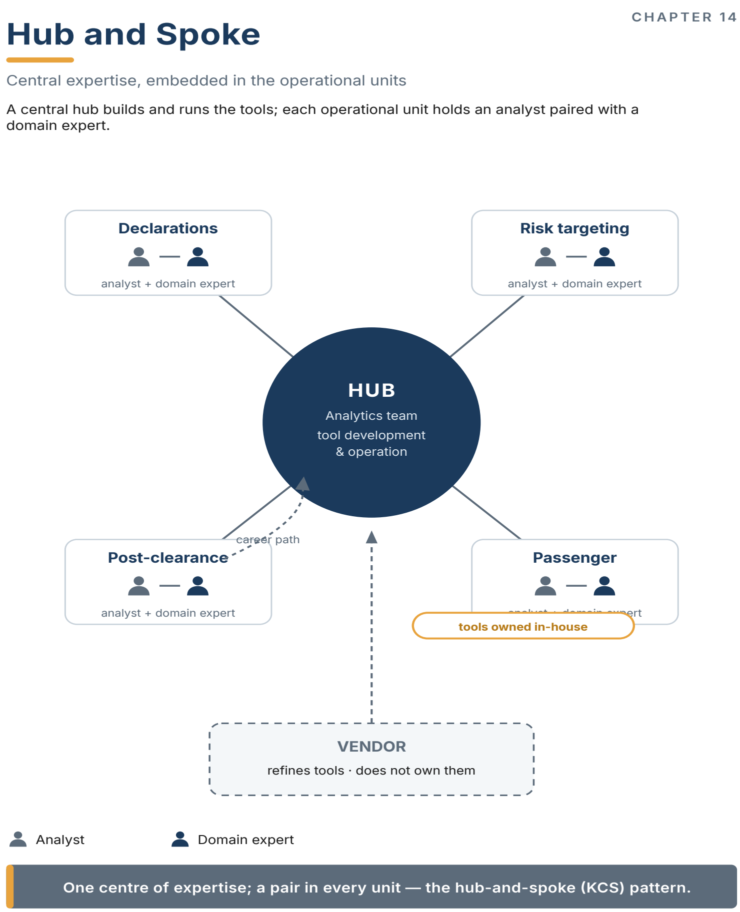


Figure 6.2 — Hub and Spoke

ers learn mandate and residual-risk decisions. Procurement/legal learn agent definition, agency, evaluation, change, audit, and exit. Engineers learn hostile content, identity, observability, and reproducible release evidence.

Capacity is a control

Set the authorised task volume against staffed review capacity. A system that creates exceptions faster than qualified people can resolve them has exceeded its safe operating envelope even if model metrics remain stable. Define maximum queue length and exception age, surge arrangements, prioritisation, and automatic demotion or fallback.

The "silicon workforce" metaphor is useful only operationally: agents need assignment, permissions, supervision, measurement, change control, and retirement. It does not make software an employee, legal person, public authority, or accountable actor.

Scale gate

Scale only when the local record answers: value captured, not merely time saved · quality stable across relevant strata · human route staffed · security exercises passed · lawful deployment unchanged · unit economics inside range · evidence export works · fallback rehearsed · owner signs residual risk. A successful demo is not on this list.

Cases as Transferable Evidence

Read the condition before the result

Cases are useful only when they answer what must be true locally. For each, record the source workflow, technology precisely identified, deployment model, measured outcome, evidence grade, elapsed institutional investment, data conditions, workforce, governance, and what is not public. Then write the local difference. The gap becomes a hypothesis and pilot design—not a reason to copy the headline number.

Korea — sequencing and compounding

Korea Customs Service publicly describes eleven production AI models, national analytics infrastructure, X-ray/manifest integration, and a later natural-language layer. It reports high-risk cargo analysis falling from roughly one hour to under a minute, e-commerce volume rising about 2.85 times without matching

Figure 7.1 — Cases at a Glance

workforce growth, and estimated annual savings near USD 92 million.

Transfer: build data access, platform, officer/analyst collaboration, and operating feedback before the generative layer; retain internal capability; communicate outcomes, not launches.

Boundary: a decade of cumulative investment, national operational data, a stable specialist cadre, research partnerships, and institutional continuity cannot be purchased in one tender. The public record does not fully disclose the governance artifacts behind every automated decision. No Korea number enters a local business case until the transfer sheet is complete.

China — image analysis at scale

China GACC reports an AI Image Analysis System across hundreds of non-intrusive inspection devices, algorithm selection, and more than 20,000 detections since 2022 with recognition accuracy above 95 percent.

Transfer: pair sensor/image data with the manifest and workflow; measure human-resource effect and false alarms; treat algorithm routing as part of the governed system.

Boundary: national scale, infrastructure, data, and legal context differ. Administration-reported outcomes do not establish a benefit range for another country.

Brazil — explainability as an operating feature

SISAM combines machine learning with human-readable reasons for targeting and has operated since 2014. Public sources report inspections around twenty times more effective than random selection.

Transfer: explanations are part of workflow acceptance and officer learning, not a compliance paragraph added after the model; compare with a real baseline.

Boundary: long-running ML targeting is evidence for disciplined analytics and explanation, not automatic proof for a GenAI agent or higher autonomy.

India, Netherlands, Uruguay, and the United States

India shows the value of codifying messy supplier identity before advanced analytics. The Netherlands shows computer vision and cooperative research routes. Uruguay shows that a smaller administration can build a focused customs analytics capability without copying the largest administrations. US CBP examples show both operational potential and why a single seizure or testimony anecdote is not a programme metric.

Across these cases, the repeatable assets are narrower than the headlines: entity resolution, data stewardship, bounded workflow choice, officer integration, evaluation, and institutional learning.

Adjacent and practice-reported evidence

Financial-crime detection offers a strong method analogy: rare adversarial events, brittle rules, network relationships, and expensive false positives. Enterprise service operations offer evidence about governed autonomy and the Operator-to-Supervisor shift.

Consultancy cases reveal how organisations redesign procurement, compliance review, and assurance.

> **ADJACENT EVIDENCE — HYPOTHESIS, NOT FORECAST.** Borrow the network method, exception design, and failure mode. Do not transfer the reported percentage into customs.

> **PRACTICE-REPORTED.** Public consultant cases can inform workflow and commercial questions. Attribute the source, retain the client and method limitations, and require a local baseline and pilot.

The cautionary museum

Recognise five exhibits:

- a launch accuracy softened by the speaker in the same sentence;
- a seizure total attributed vaguely to “AI and big data”;
- an “up to” percentage with no primary evaluation;
- a real single-window or process gain relabelled as AI;
- a pilot that passes curated cases but never secures data, ownership, security approval, operating budget, or a staffed exception route.

The fifth is especially important. A proof of concept can be technically successful and institutionally negative. If there is no conversion path, the programme tested the model before it tested the administration.

The transfer sheet

For every external case, complete: source and date · named workflow · technology · evidence grade · metric denominator · starting baseline · elapsed investment · data and legal conditions · workforce · governance · unknowns · local differences · local hypothesis · pilot threshold · expiry date. An incomplete sheet keeps the case out of the financial model.

What to Do on Monday

The first ninety days

The first quarter should create decision-grade evidence and shared infrastructure, not a large platform commitment.

Days 1–15 — establish command. Name the executive owner and a small cross-functional gate team: operations, legal/privacy, security, data/IT, procurement, workforce, and finance. Inventory live and experimental AI, including personal subscriptions and API keys. Stop new sensitive data from entering unapproved paths and provide a sanctioned synthetic/public- data sandbox. Select one bounded case loop and measure its current volume, time, quality, exceptions, cost, and owner.

Days 16–30 — define the task. Complete the decision card, Autonomy Ladder rung, agency profile, data-flow/residency map, applicability record, affected-party route, initial value bridge, and demotion/fallback design. Choose the approved corpus and feeds. Name their custodians and freshness rules.

Days 31–45 — prepare evidence. Build a representative evaluation set with hard, multilingual, incomplete, duplicate, and hostile cases. Keep a sealed holdout. Define baseline, repeated trials, error taxonomy, abstention, latency, cost, security, human-queue, and complete-case thresholds before testing.

Days 46–60 — buy or build visibly. Run the twenty-case prequalification where a vendor is involved. Require the six-question response, lawful deployment, component and subprocessor register, identity/tool diagram, trajectory export, change matrix, incident cooperation, commercial normalisation, knowledge transfer, and exit rehearsal. Start the evidence room.

Days 61–75 — run shadow mode. Compare with the current workflow. Sample trajectories. Measure correction and override reasons, state-dependent failure, unit economics, exception capacity, and adoption. Run hostile- document, excessive-authority, stale/conflicting-source, and freeze-preserve-fallback exercises.

Days 76–90 — make a gate decision. Publish a short memo: evidence, threshold result, residual risk, control gaps, value range, legal/security status, human capacity, and recommendation. The options are promote to a limited live pilot, hold and tune, demote/simplify, or stop. A stopped pilot that prevents an unsuitable production system is a successful governance outcome.

The board dashboard

Keep it decision-oriented:

Question	Minimum evidence
Is value being captured?	Baseline, completed-case volume, unit cost, named conversion action
Is quality acceptable?	Strata, repeats, complete-case success, overrides, abstentions, adverse burden
Is authority contained?	Current mandate, rung, agency profile, denied-tool test, demotion record
Is the human route viable?	Queue, exception age, staffing, calibration, stop authority
Is the service lawful and sovereign?	Applicability record, approved data path, providers/subprocessors, next review
Can we investigate and exit?	Exportable trajectory/evidence room, incident exercise, fallback and exit rehearsal

The 12–24 month watch

For capability, cost, sovereignty, trader-side agents, standards, and law, record probability, earliest plausible decision date, official source, no-regret preparation, irreversible commitment to avoid, readiness, owner, and next review. Do not buy against a forecast. Build options that remain useful if the forecast is wrong.

Twelve non-negotiables

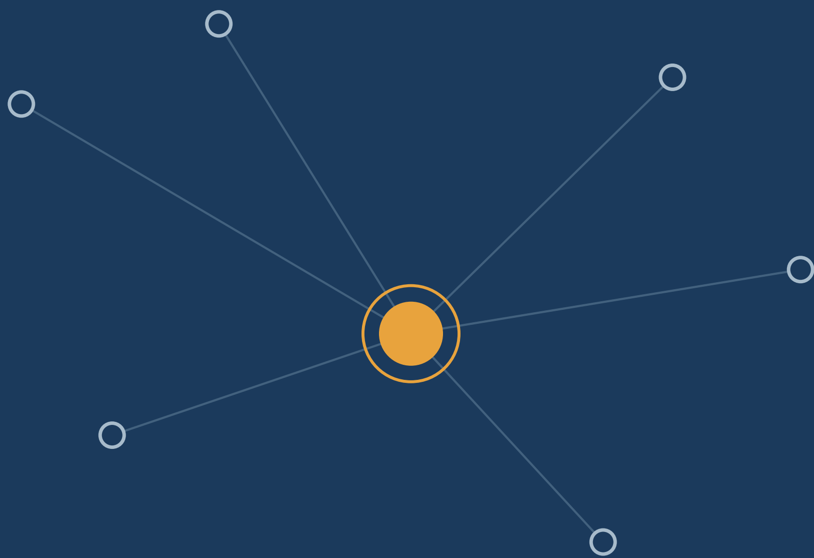
1. One task, one baseline, one accountable owner.
2. Fixed pipeline and adaptive agent are named accurately.
3. Autonomy and agency are recorded separately.
4. Decisions transfer one rung at a time; accountability never does.
5. No sensitive data in consumer subscriptions or unreviewed APIs.
6. A commercial region is not sovereign accreditation.
7. Twenty cases pre-qualify; they do not prove production quality.
8. Adjacent and practice-reported numbers keep their labels.
9. Every consequential task has a mandate, internal safety case, human route, demotion trigger, and fallback.
10. The administration owns evaluation and an exportable evidence room.
11. Material model, prompt, corpus, tool, identity, region, and dependency changes are governed releases.
12. Renewal is a new decision: continue, optimise, compete, migrate, demote, or retire.

Final question

The mandate of customs remains collect the revenue, protect the border, and facilitate trade. Agentic AI changes the available operating model, not the institution's responsibility. The durable executive question is:

> ***What may this system decide, what can it technically do, who signed for it, what evidence earned that authority, and how will we know when to take it back?***

If the leadership team can answer that question per task, it is no longer being led by a technology category or vendor narrative. It is running an AI programme.



A DECISION BOOK FOR BORDER LEADERS

AI Agents at the Border is a practical guide for leaders who must decide what may be delegated — and what must remain human.

A field guide to buying, governing and operating
AI that can act in trade and customs environments.

> **DECISIONS MAY TRANSFER WITHIN A SIGNED MANDATE.**
> **ACCOUNTABILITY DOES NOT.**

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